

Artificial Intelligence Literacy and Pedagogical Use of AI among Sri Lankan School Teachers: Implications for Digital Education

F. M. Nawastheen, N. G. L. S. J. Liyanage, K. S. P. Sisindra & S. Shifaan

The Open University of Sri Lanka | National Institute of Education, Sri Lanka

ICODE2026

UNIVERSITY OF MORATUWA
Centre for Open and Distance Learning

Introduction

Artificial Intelligence (AI) is significantly transforming teaching and learning environments globally.

Teachers increasingly use AI for:

- Lesson planning and structure
- Generating teaching materials
- Conducting assessments
- Providing student feedback
- Facilitating personalised learning

Effective classroom integration depends on teachers' AI literacy — now an essential professional competency for modern educators.

Research Problem

ICODE 2026 - Academic Research Conference - Session B

Current Research Focus

AI is increasingly being used by Sri Lankan teachers for lesson planning, assessment, and content preparation.:

- Rapid growth in AI use in schools
- Limited institutional guidance
- Teachers rely on personal experience
- Classroom practices vary considerably

The Practical Challenge

As AI use expands,

- teachers' AI literacy remains unclear,
- classroom use is not well understood, and
- policymakers lack national evidence to guide teacher development.

Understanding teachers' AI literacy and pedagogical use is essential for developing evidence-based teacher education and AI policy in Sri Lanka.

Aim and Objectives

STUDY AIM

To examine AI literacy and the pedagogical use of AI among Sri Lankan school teachers.

Specific Objectives

1. Identify the specific AI tools being used by teachers
2. Assess the current level of AI literacy among the target population
3. Examine the extent of pedagogical use of AI in classrooms
4. Investigate the relationship between AI literacy and pedagogical use

Literature Review

AI in Education

AI provides robust support for modern educational practice, specifically assisting teachers with:

- Efficient lesson planning
- Automated and formative assessment
- Generating diverse learning resources
- Providing timely, constructive feedback
- Designing personalised learning pathways

AI Literacy

A multifaceted competency. Key dimensions necessary for educators include:

- Foundations: basic concepts & mechanisms
- Application: practical use in scenarios
- Critical Evaluation: assessing outputs & bias
- Ethics & Privacy: responsible AI use
- Human–AI Interaction: collaborative synergy
- Societal Awareness: broader impacts

Research Gap

Previous studies globally have reported:

- Moderate overall AI literacy among educators
- A rapidly increasing rate of AI adoption in schools

However, critical gaps remain:

- Limited empirical evidence linking AI literacy directly with pedagogical use
- Very limited evidence reflecting the context of Sri Lankan schools

This study contributes essential empirical evidence to address this localized and conceptual gap.

Methodology Overview

1	**Research Approach**	Quantitative approach to allow for statistical measurement and generalization.
2	**Research Design**	Survey research design to capture cross-sectional data from a wide geographic area.
3	**Study Area**	Conducted nationwide across Sri Lanka.
4	**Instrument**	Structured online questionnaire designed specifically for educators.
5	**Data Collection**	Administered digitally via Google Forms for efficient distribution and response tracking.
6	**Data Analysis**	Frequency, Mean & Standard Deviation, Pearson's correlation

Population and Sampling

Sampling Technique: Non-probability snowball sampling

Target Population

Practicing school teachers across the Sri Lankan education system.

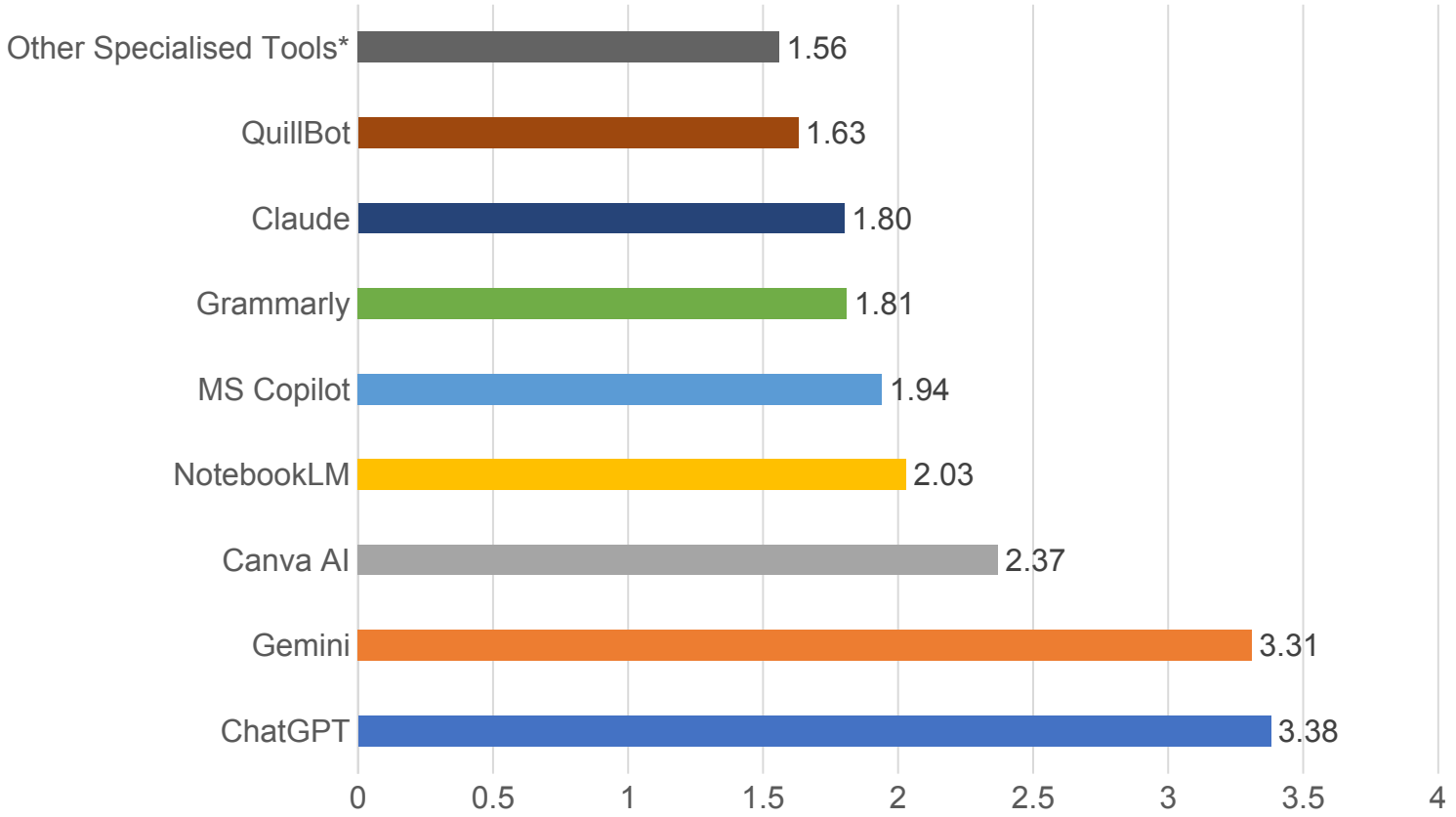
Sample Size

The final effective sample size achieved for this research was N = 410.

Coverage

Responses collected from all 25 districts, encompassing Sinhala, Tamil, and English medium schools.

Findings: AI Tools Used



Interpretation

Teachers mainly rely on general-purpose generative AI (ChatGPT, Gemini) rather than specialised educational AI tools designed specifically for pedagogical tasks.

**Other Specialised Tools = average of MagicSchool.ai, Eduaide.ai, Wolfram Alpha, Photomath, and Perplexity AI*

Findings: AI Literacy

AI Literacy Dimension	Mean (SD)
Societal Awareness	3.87 (1.02)
AI Foundations	3.61 (0.84)
Human–AI Interaction	3.55 (1.08)
AI Use & Application	3.54 (1.00)
Ethics & Privacy	3.54 (1.05)
Critical Evaluation	3.49 (0.99)

5-point scale · N = 410

Interpretation

- Teachers demonstrated moderately high AI literacy overall.
- Confidence in evaluating AI-generated content (Critical Evaluation) was comparatively lower.

Findings: Pedagogical Use

3.18

Frequency of AI Use
(Mean Score)

3.33

Pedagogical Application
(Mean Score)

Interpretation

Teachers have begun integrating AI into their teaching practices, but usage indicates AI has not yet become routine, everyday classroom practice.

Relationship Analysis

Relationship between AI Literacy and Pedagogical Use

AI Literacy Dimension → Pedagogical Application	Pearson's r
AI Use & Application	.737
Critical Evaluation	.605
Ethics & Privacy	.598
AI Foundations	.583
Human–AI Interaction	.569
Societal Awareness	.511

All correlations were positive and statistically significant (p < .01). Practical application skills most strongly predict classroom integration.

Discussion

1. Teachers strongly prefer widely available, general AI tools over niche educational software
2. AI literacy among the sampled teachers is generally high across most dimensions
3. Actual classroom integration remains moderate, indicating a transition phase
4. Critical evaluation skills concerning AI outputs require targeted strengthening
5. AI literacy positively and significantly influences the extent of pedagogical use
6. These findings align consistently with observations from previous international studies

Practical Implications

Teacher Education

Curricula must fundamentally strengthen AI literacy, preparing pre-service teachers for an AI-augmented educational landscape.

Professional Development

In-service training should pivot towards teaching practical AI skills, enhancing critical evaluation, and navigating ethical AI use.

Educational Policy

Policymakers need to establish institutional AI guidelines and facilitate greater access to specialised educational AI applications.

Recommendations

Strategic & Institutional

- Policy Level: develop clear, actionable institutional AI guidelines to govern responsible usage in schools.
- Institutional Level: actively increase awareness and procurement of specialised, purpose-built educational AI tools.

Practical & Future Scope

- Practitioners: commit to strengthening continuous AI literacy through ongoing professional development.

Future Research Should Focus On:

- Probability sampling for wider generalization
- The impact of institutional support
- Subject-specific pedagogical investigations
- Digital infrastructure readiness studies

Conclusion

1. Teachers demonstrated moderately high AI literacy levels across core dimensions
2. Despite this literacy, day-to-day pedagogical use of AI remains moderate
3. Practical AI competencies were identified as the strongest predictor of successful classroom integration
4. Strengthening AI literacy will directly support responsible and meaningful AI integration in Sri Lankan schools

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Thank you for your attention.